

Mehak Khan

Electrical Engineering · Analog / Mixed-Signal IC Design

Third-year BAsC Honours, University of Waterloo · Graduating April 2028
President's Scholarship of Distinction

[LinkedIn](#) · [Email](#) · [Resume](#)

About

I work across the stack from transistor-level analog IC design up through digital hardware and, most recently, into the ML side of hardware-software co-design. My projects lean toward proving things out from first principles — measured device characterization over textbook constants, real debugging over polished demos, and being upfront about what's finished versus still in progress.

Currently targeting Winter 2027 co-ops in analog/mixed-signal design and verification.

Skills & Tools

Analog / IC Design	Cadence Virtuoso, Xschem, ngspice, SKY130 PDK, subthreshold/analog circuit design, bandgap references, error amplifiers
Digital Hardware	SystemVerilog, Vivado, Vitis, FPGA timing closure and pipelining, FSM design
PCB Design	Altium, KiCad, EasyEDA, signal integrity, power management circuits
Software / ML	Python, PyTorch, snnTorch, NeuroKit2, hardware-aware quantization
Industry	IEC 61850, protection relay testing, UPS/power electronics validation, power quality analysis

Experience

Technip Energies (Qatar)	Co-op, Jan–Apr 2026 — protection relay testing, IEC 61850, UPS validation, HV switchgear commissioning
LCRAIN Lab, York University	SNN (spiking neural network) research
Toyota Canada	Software co-op — cloud/backend
Landscape Direct Corps	Software co-op — backend

Pipelined Matrix-Vector Multiplication Engine

Course: ECE 327 — Digital Hardware Systems, University of Waterloo · **Team:** Mehak Khan & Hiya Patel · **Platform:** AMD Kria KV260 (Zynq UltraScale+)

Tools: SystemVerilog, Vivado/Vitis, XSim · **Repo:** [fpga-mvm-engine](https://github.com/mehak-khan/fpga-mvm-engine)

Designed and implemented a parameterized FPGA matrix-vector multiplication accelerator built from parallel 8-element dot-product engines, per-lane accumulators, SRAM-backed matrix/vector storage, and an FSM controller. The main challenge was keeping memory reads, pipeline-valid signals, and accumulation boundaries aligned across a deeply pipelined multi-lane datapath.

Architecture

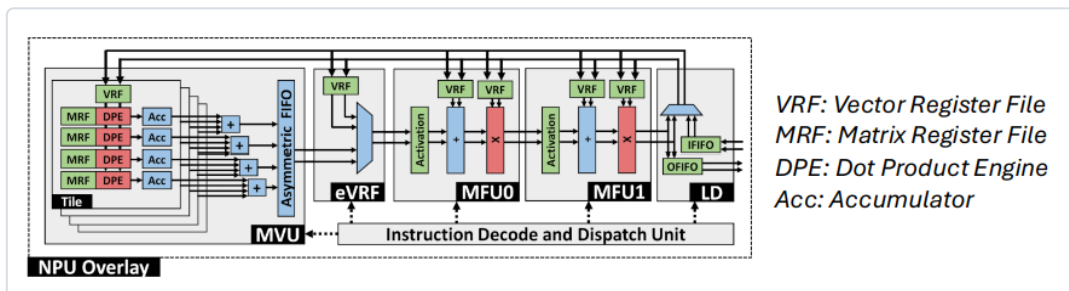
Each output lane pairs a matrix memory with an 8-element dot-product engine (`dot8`) and an accumulator; a shared vector memory feeds every lane. The dot-product unit computes eight parallel multiplications, then reduces them through a balanced $4 \rightarrow 2 \rightarrow 1$ adder tree across a six-cycle registered pipeline, with the valid signal delayed in lockstep. The number of output lanes (`NUM_OLANES`) is a synthesis-time parameter, so parallelism scales without redesigning the datapath. A two-state FSM controller generates matrix/vector read addresses incrementally (avoiding a per-cycle multiply for row addressing) and produces the `first` / `last` accumulation boundaries.

Hardest Bug

The controller-issued control signals (`first` , `last` , `valid`) don't reach the accumulator on the same cycle they're issued — they need to be delayed by 7 cycles (1 memory-read cycle + 6 dot-product pipeline stages) to line up with the arithmetic result they describe. Getting this alignment right, and debugging an integration-level signal-name mismatch that caused output to diverge only partway through computation, was the core system-level challenge.

Metric	Result
Critical path delay (post-implementation)	2.621 ns
Target clock frequency	> 350 MHz
LUT / LUTRAM / FF / BRAM utilization	~5% / ~1% / ~1% / ~6%

What I learned: high-frequency RTL isn't just registers on arithmetic — data movement, control movement, memory latency, and arithmetic latency all have to be treated as one timing problem.



Reference NPU/MVM datapath (Microsoft BrainWave-style) used as the system-level model this project's lane/accumulator architecture was adapted from — not my own diagram.

PDK-Agnostic Subthreshold Bandgap Voltage Reference

Project: UWASIC (Analog & Systems IC Design Team) — Tiny Tapeout submission · **Team:** Mehak Khan, Zoey Situ, review from Henry Wu

Tools: Xschem, ngspice, SKY130 PDK, Verilog-A / OpenVAF

A MOSFET-only (no BJT) subthreshold bandgap reference targeting -40°C to 120°C in SKY130, sized from measured device characterization rather than textbook constants so the methodology can migrate across CMOS processes. The reference combines a CTAT gate-source voltage with a PTAT ΔV_{GS} term from differently-sized subthreshold MOSFETs, weighted by resistor ratios to cancel first-order temperature dependence.

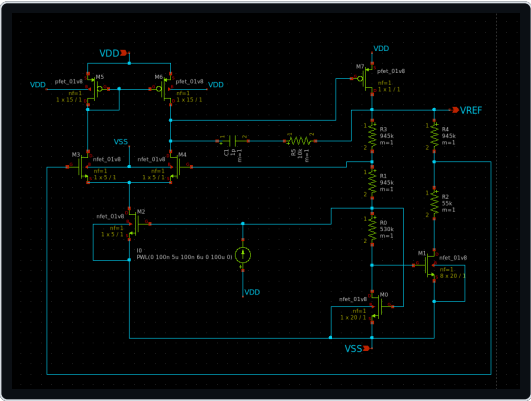
Key Debug

The first working implementation showed excessive PTAT contribution. Increasing `R0` alone didn't fix it — the real issue was in the topology. I removed PMOS device `M7` and re-derived resistor values from measured SKY130 subthreshold parameters ($\eta \approx 1.48$, CTAT slope $\approx -0.63 \text{ mV}/^{\circ}\text{C}$) instead of the original reference design's process constants.

Metric	Result
Nominal V_{REF}	$\sim 630\text{--}633 \text{ mV}$
Temperature coefficient	$\sim 89 \text{ ppm}/^{\circ}\text{C}$ ($\Delta V_{\text{REF}} \approx 9 \text{ mV}$, -40°C to 120°C)
Low-frequency PSRR	$\sim 65 \text{ dB}$ (target was $\sim 40 \text{ dB}$)
Error-amp gain / unity-gain freq.	$\sim 41 \text{ dB}$ / $\sim 10 \text{ MHz}$

Transistor-level AC analysis also revealed frequency-dependent PSRR degradation not captured by the simplified Verilog-A behavioral model — about $+2 \text{ dB}$ of supply-to-output peaking around $100\text{--}150 \text{ kHz}$, traced to error-amplifier/compensation loop dynamics.

Status: transistor-level BGR operational, Verilog-A model validated against it; final PSRR tuning and physical-budget verification in progress.



Final transistor-level MOSFET subthreshold bandgap reference, simulated in Xschem/ngspice on the SKY130 PDK.

Neuromorphic Stress-Responsive Wearable

Long-term goal: a discreet wearable that detects sustained physiological stress and responds with a private, heartbeat-like haptic cue at the wrist · **Tools:** Python, PyTorch, snnTorch, NeuroKit2, Xschem, ngspice, SKY130

Repo: [CPTSD-Wearable](#)

Phase 1 asks: can a physiological stress classifier be trained in software and then have its real, quantized weights mapped onto a physically characterized CMOS analog-compute primitive? The chain: WESAD data → features → spike encoding → 4-bit SNN → trained weights → measured SKY130 conductances → required gate biases → transistor-level analog current summation.

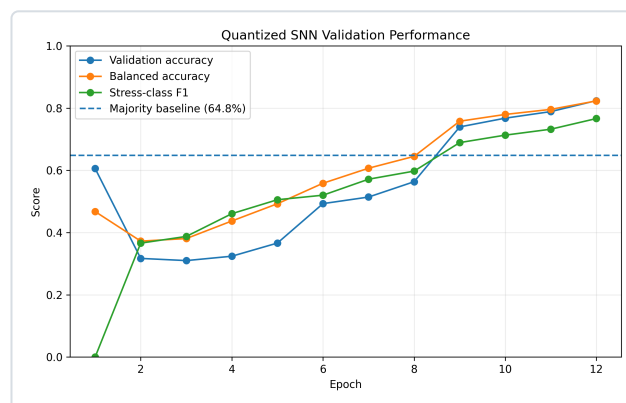
Getting From a Broken Classifier to a Working One

Early SNN versions collapsed to predicting the majority class (baseline) — raw accuracy looked fine, but stress recall and F1 were 0%. Adding firing-rate telemetry and balanced-accuracy/F1 tracking surfaced this, and switching from raw ECG/EDA traces to six NeuroKit2-derived physiological features (heart-rate stats + phasic EDA stats) fixed the class separability problem.

Metric	Result
Validation accuracy / balanced accuracy	82.4% / 82.3%
Stress recall / F1 (majority-class baseline: 64.8%)	82.0% / 76.6%
Two-cell analog current-sum vs. ideal model	~17.8% magnitude discrepancy

Trained 4-bit weights were mapped to measured SKY130 NMOS conductances (a 5-point $G(V_{GS})$ lookup table), then to required gate biases, and validated with a two-transistor analog current-summing circuit using biases derived from real trained weights — demonstrating the full software-to-hardware mapping, not just the algorithm in isolation.

Honest scope: this is a proof of concept on two WESAD subjects with window-random (not subject-held-out) validation — not a clinical or diagnostic result. Sensor hardware and the haptic subsystem are future phases.



Validation performance of the 4-bit quantized SNN across training epochs, versus the 64.8% majority-class baseline.

Watolink — Main PCB for an EMG-Controlled Prosthetic Hand

Role: Hardware Lead · **Team:** Mehak Khan, Fiona, Aidan

Tools: EasyEDA, ESP32-S3, ADS1115, mixed-signal 2-layer PCB design

Designed the central control and power PCB for a wearable EMG-controlled prosthetic hand: battery charging, 6 V servo power, 3.3 V digital power, an ESP32-S3, dual ADS1115 ADCs, six servo interfaces, force-sensor acquisition, and I²C — all on one 2-layer board.

Engineering Problem 1 — Designing for Manufacturability

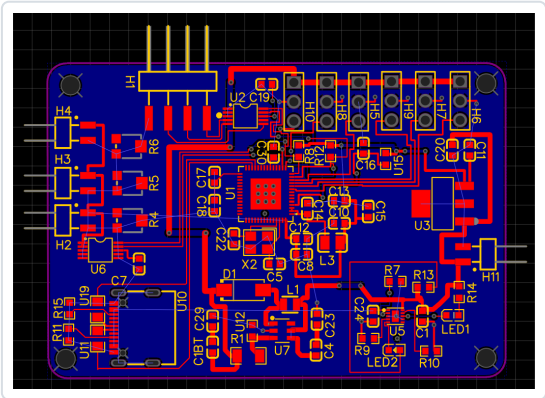
The originally specified battery charger (BQ2560-family) used a package too fine-pitch to route on a 2-layer board. Rather than add PCB layers or force an unreliable layout, I swapped to the **BQ24040**, which met the same charging requirements in a far more practical package — a reminder that electrical spec alone doesn't make a good part choice.

Engineering Problem 2 — EMG Signal Integrity

Measured EMG data was noisy and inaccurate. Rather than adding redundant sensing channels, I traced the likely cause to **electrode-to-skin impedance mismatch**: unequal source impedances at the two differential inputs convert common-mode interference into a differential error the instrumentation amp can't reject. The fix under development is a high-input-impedance unity-gain buffer at each electrode, isolating the amplifier from the variable electrode impedance rather than trying to passively match it.

Board	Detail
Domains	3.3 V logic (ESP32-S3, ADCs, sensing) / 6 V switching supply (servos), isolated
Sensing	2× ADS1115 (16-bit) — 3 force-sensitive resistors + servo feedback, shared I ² C
Actuation	6 PWM channels — wrist, thumb flexion/opposition, index, remaining fingers

Status: main PCB designed and reviewed; fabrication and bring-up pending, buffered EMG front-end still to be prototyped and validated against the unbuffered baseline.



Completed 2-layer Main PCB layout — top copper in red, bottom copper in blue.